Course- B.Tech
Course Code- CSET343
Year- 4TH Year

Type- Specialization Elective
Course Name- AI in Healthcare
Semester- VII

Lab Assignment -3

Objective: Agenda: Perform data acquisition, cleaning, preprocessing and analysis on Tabular, Textual, Image, and Signal medical health data.

- Acquire and inspect real-world medical datasets across four different modalities.
- Apply modality-specific cleaning techniques, including healthcare-specific concerns like implausible clinical values, PHI (Protected Health Information) removal, and imaging metadata anonymization.
- Preprocess each data type into a model-ready format (scaled tensors, token sequences, image tensors, signal windows).
- Understand why healthcare data requires extra rigor compared to generic datasets (missingness encoded as valid-looking values, class imbalance, privacy constraints, signal noise).

Part	Data Type	Dataset	Link
A	Tabular	Pima Indians Diabetes Dataset	Pima Indians Diabetes
B	Textual	MTSamples Medical Transcriptions	MT Samples
C	Image	Chest X-Ray (Pneumonia) Dataset	Chest X-Ray
D	Signal	MIT-BIH Arrhythmia Database (ECG)	MIT-BIH Arrhythmia

Part A

- **Data Acquisition & Initial Inspection** Load the CSV, print shape, dtypes, and summary statistics.
- **Data Cleaning** Treat implausible zeros as missing values and impute them appropriately. Remove exact duplicate rows.
- **Data Preprocessing** Scale numerical features and confirm the target is properly encoded.
- **Inferential Statistics.** Use Chi-square Test, Hypothesis Test (T-test) to draw some inferences from the dataset.
- **Exploratory Data Analysis** Visualize feature distributions, class balance, and the correlation between features and the outcome.
- **Feature Engineering & Selection** Create at least one derived clinical feature (e.g., a BMI category) and select the most predictive features using a statistical or model-based methods.

Part B

- **Data Acquisition and analysis.** Load the dataset, inspect he transcription and specialty columns, and check for missing/empty reports.
- **De-identification / PHI Check.** Scan the text for residual patient identifiers (names, dates, MRNs, phone numbers) using regular expression, and redact any.
- **Text Cleaning.** Remove HTML artifacts, section header noise, extra whitespace, and non-alphanumeric characters.
- **Tokenization & Stop word Removal** Tokenize the cleaned text and remove generic stop words.
- **Normalization** Lemmatize tokens and expand common medical abbreviations
- **Feature Extraction** Convert the cleaned, tokenized reports into numerical features using TF-IDF

Part C

- **Data Acquisition & Inspection** Load the folder structure, count images per class/split, and inspect image dimensions and file formats.
- **Metadata Handling & Anonymization** For DICOM-based datasets, strip patient-identifying metadata tags before any further processing.
- **Image Quality Check & Cleaning** Detect and remove corrupted, unreadable, or near-duplicate images.
- **Preprocessing** Resize, convert to a consistent channel format, and normalize pixel intensities.
- **Data Augmentation** Apply augmentations appropriate to X-rays (rotation, flips, brightness/contrast) to counter class imbalance and improve generalization.
- **Exploratory Data Analysis** Visualize sample images per class and quantify class imbalance.

Part D

- **Data Acquisition** Load a record using wfdb, inspect the raw signal, sampling rate, and annotation file.
- **Signal Inspection & Cleaning** Check for flatlines, clipped values, or missing samples in the raw signal.
- **Noise Removal / Filtering** Remove baseline wander and powerline interference using bandpass and notch filters.
- **Segmentation** Detect R-peaks and segment the continuous signal into individual heartbeat windows.
- **Feature Extraction** Compute time-domain and frequency-domain features (e.g., RR-interval statistics, heart rate variability, spectral power).
- **Normalization** Normalize each beat window (and/or extracted feature vector) to a consistent scale.